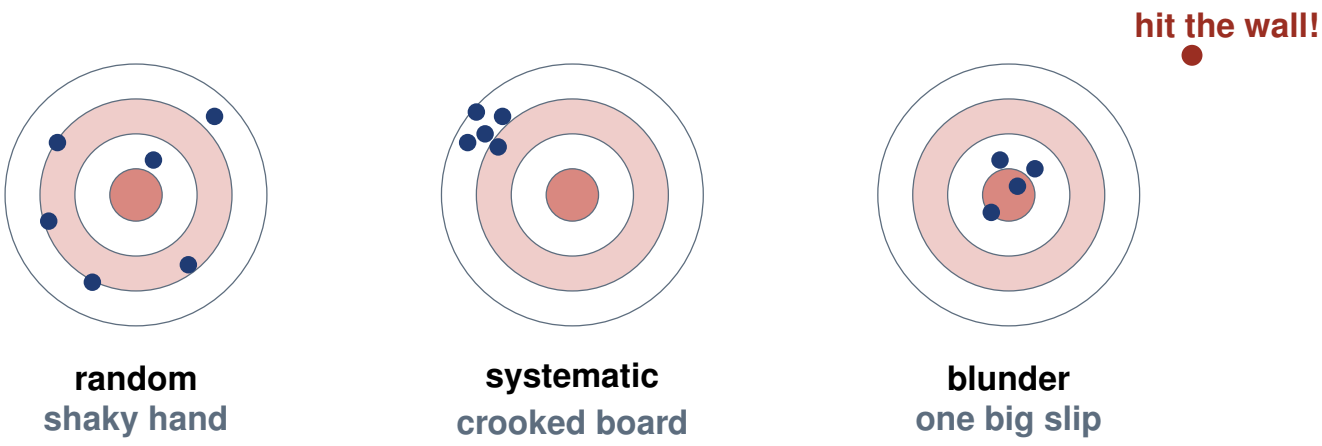
Error Propagation and Error Budgets

BIG IDEA

Every small mistake adds a little, and we add them up to see the whole.



THE IDEA

Throw darts. If they land all over, that is **random error**: your hand shakes a little differently each time. If they land in a tight group but always to one side, that is **systematic error**: maybe the board is crooked. If one dart hits the wall, that is a **blunder**. Each one is fixed in a different way.

REMEMBER

- ✓ Practice shrinks random error. Fixing the cause removes systematic error. Blunders are thrown out.
- ✓ An **error budget** lists every little thing that can move a dot, like packing a backpack.
- ✓ Averaging many tries shrinks random error, but it cannot fix a crooked board.

TRY THIS

Throw five paper balls at a target. Then tilt the target a little and throw five more. Which group is spread out, and which is shifted?

The full module is in the Reference Book.